

Exercise 1

Computer simulation is a powerful tool in the engineering design process. Starting from design specifications, an initial design typically includes selection of system and circuit configurations, as well as component types and values. In this experiment, component and system models are constructed by MATLAB/SIMULINK. These models are the basis upon the designer can choose parameter values. One must take into account the fact that actual parameter values will not match their nominal values because of inevitable production tolerances, changes in environmental conditions (such as temperature), and aging. In the design verification step, worst-case analysis (or other reliability and production yield analysis) is performed to judge whether the specifications are met under all conditions, i.e., for expected ranges of component parameter values. Computer simulation is very well suited for this task: using reliable models and appropriate simulation setups, the system performance can be tested for various sets of component parameter values. One can then perform design iterations until the worst-case behavior meets specifications, or until the system reliability and production yield are acceptably high.

In the design verification of power electronic systems by simulation, it is often necessary to use component and system models of various levels of complexity:

1. *Detailed, complex models that attempt to accurately represent physical behavior of devices.* Such models are necessary for tasks that involve

finding switching times, details of switching transitions and switching loss mechanisms, or instantaneous voltage and current stresses. MATLAB/SIMULIN provides libraries of such device models. To complete a detailed circuit model. With fast-switching power semiconductors, simulation time steps of a few nanoseconds or less may be required, especially during on/off switching transitions. Because of the complexity of detailed device models, and the fine time resolution, the simulation tasks can be very time consuming. In practice, time-domain simulations using detailed device models are usually performed only on selected parts of the system, and over short time intervals involving a few switching cycles at most. Devices for power converters, and detailed physical device modeling, are areas of active research and development beyond the scope of this practical.

2. *Simplified device models.* Since an on/off switching transition usually takes a small fraction of a switching cycle, the basic operation of switching power converters can be explained using simplified, idealized device models. For example, a MOSFET can be modeled as a switch with a small (ideally zero) on-resistance R_{on} when on, and a very large off-resistance (ideally an open circuit) when off. Such simplified models yield physical insight into the basic operation of switching power converters, and provide the starting point for developments of analytical models described throughout this experiment. Simplified device models are also useful for time-domain simulations aimed at verifying converter and controller operation, switching ripples, current and voltage stresses, and responses to

load or input transients. Since device models are simple, and details of switching transitions are ignored, tasks that require simulations over many switching cycles can be completed efficiently using general-purpose circuit simulators. In addition, specialized tools have been developed to support fast transient simulation of switching power converters based on idealized, piecewise-linear device models, or a combination of piecewise-linear and nonlinear models.

1: Buck Converter

This exercise shows how to model a switching power supply that converts a 12V DC supply into a regulated 5V DC supply. The model can be used to both size the inductance L and smoothing capacitor C . By selecting between continuous and discrete controllers, the impact of discretization can be explored.

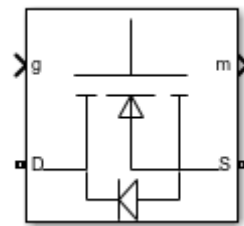
Modeling the switching devices as MOSFETs rather than ideal switches ensures that device on-resistances are correctly represented. The model also captures the switch-on/switch-off timing of the devices, this depending primarily on the gate capacitance values and the PWM driver output resistance. The model here can be used to determine the on-resistance values required for the ideal switches, plus timing offsets if necessary. Using the ideal switching approach of Simscape Power Systems in Simulink can be used to simulate more complex power converters.

1) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Electrical Sources. Double click the icon, you can change the

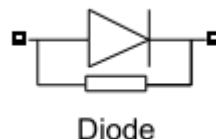
parameters.



2) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Power Electricals. Double click the icon, you can change the parameters (ignore the m terminal).



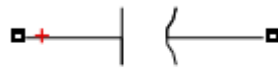
3) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Power Electricals. Double click the icon, you can change the parameters.



4) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Elements. Double click the icon, you can change the parameters.



5) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Elements. Double click the icon, you can change the parameters.



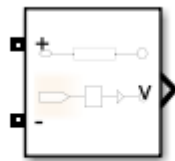
Series RLC Branch

- 6) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Elements. Double click the icon, you can change the parameters.



Series RLC Branch

- 7) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Measurements. Double click the icon, you can change the parameters.



Voltage Measurement

- 8) Simulink → Sources. Double click the icon, you can change the parameters.



Pulse
Generator

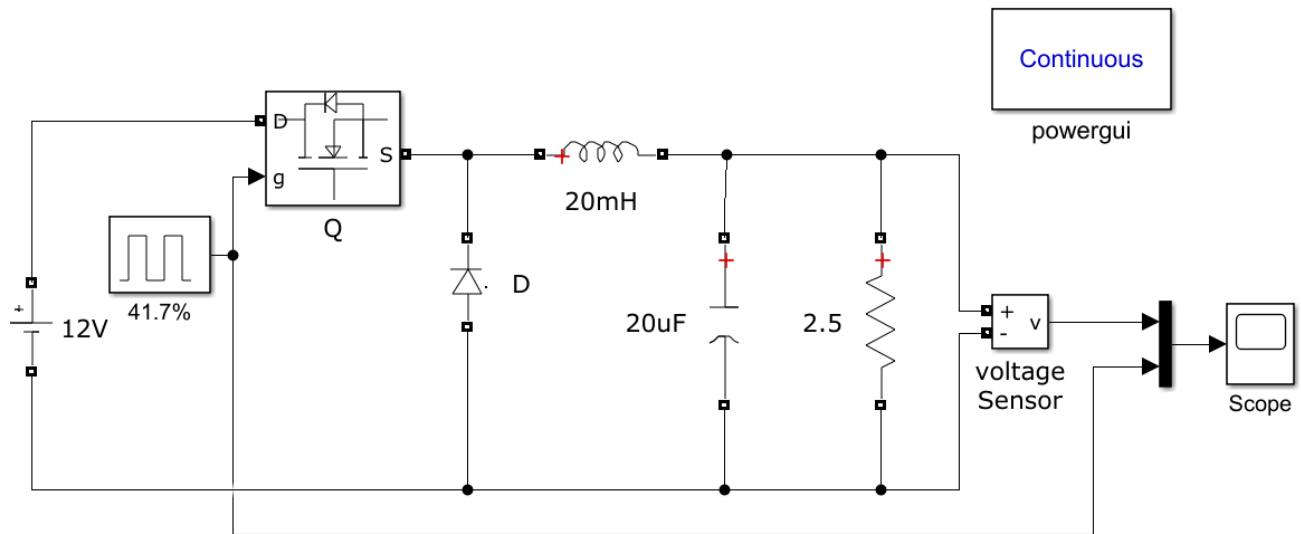
- 9) Simscape → Electrical → Specialized Power Systems → Fundamental Blocks.



powergui

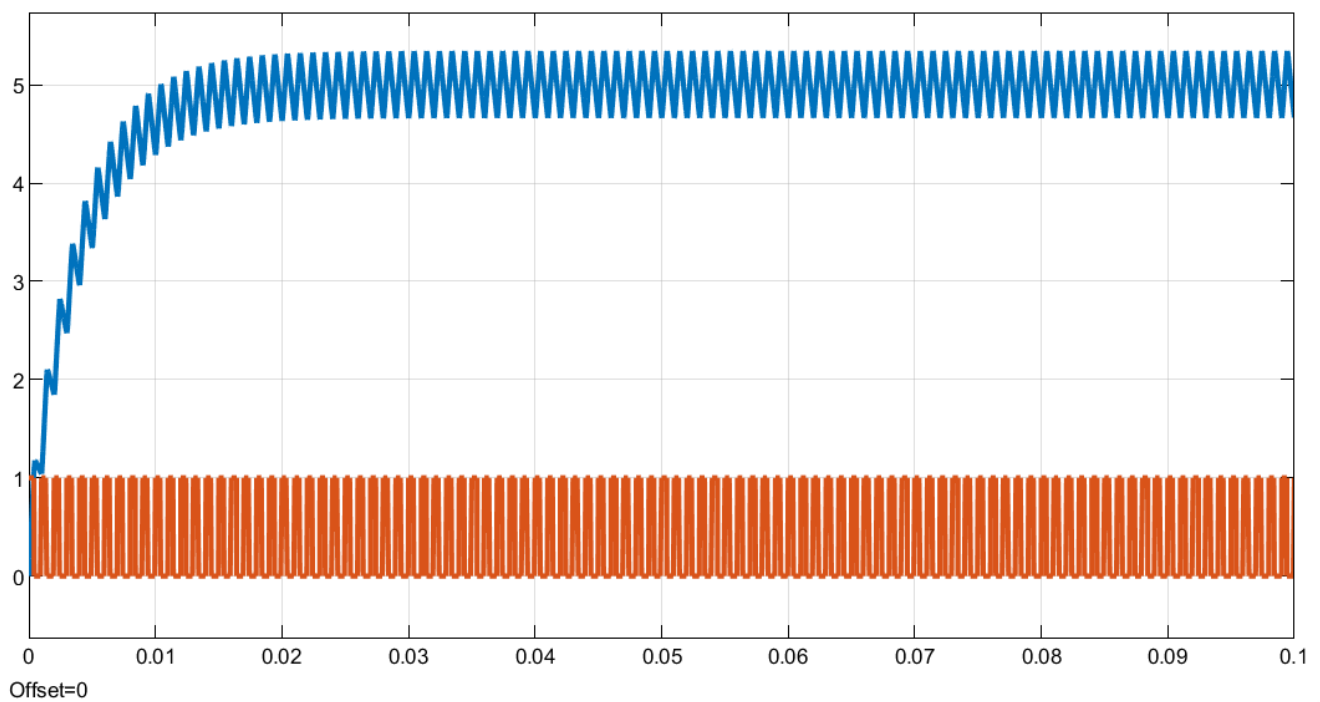
- 10) Connect the above parts like this. You can change the parts of value to

get different result.

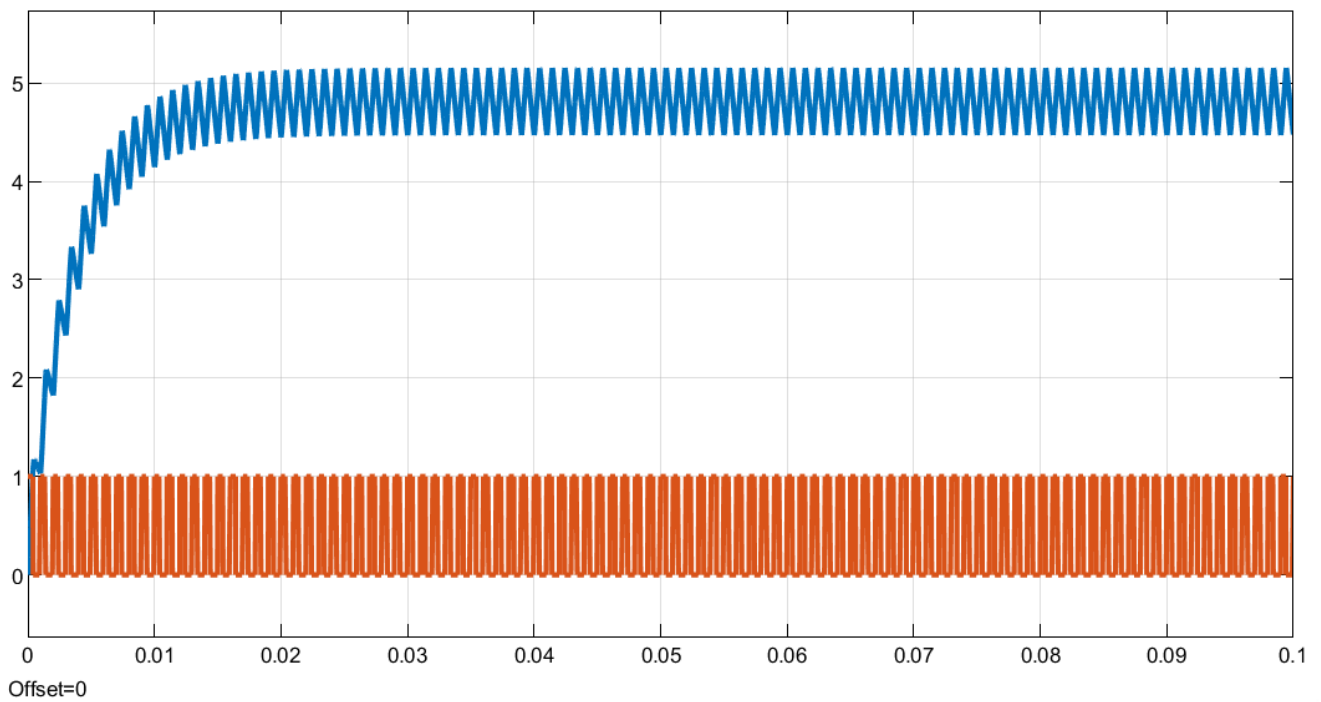
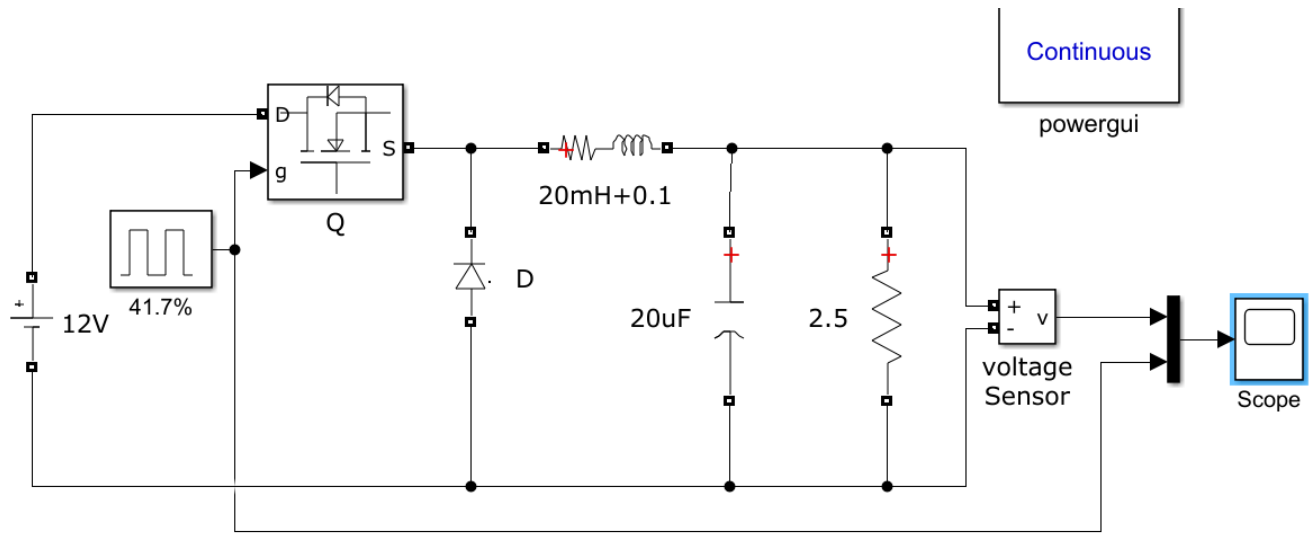


11) Simulation Results

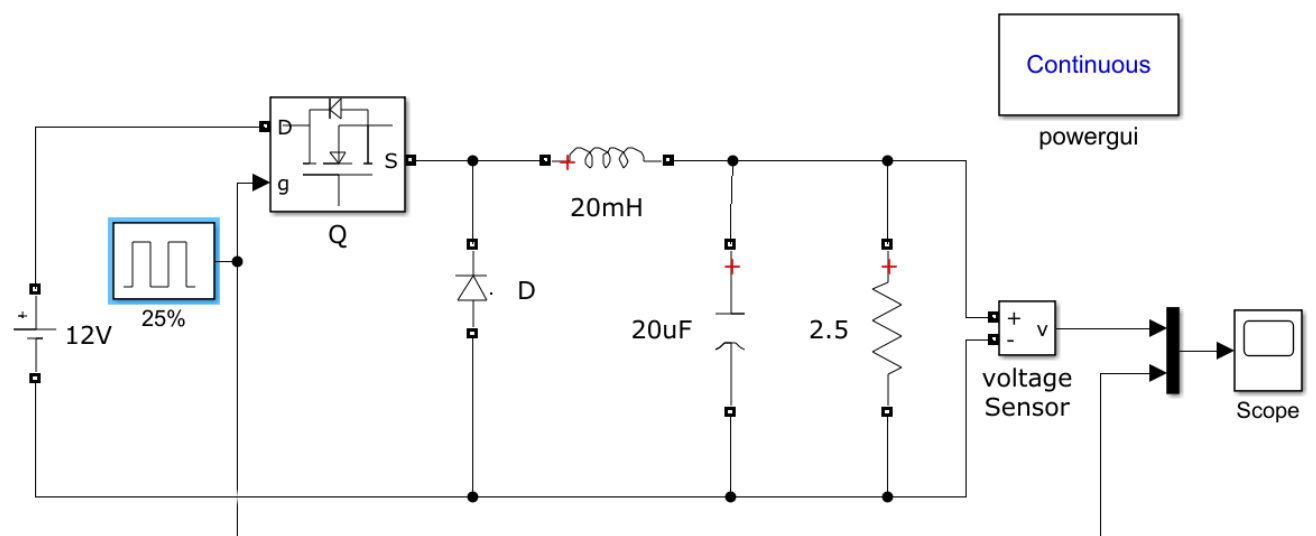
The plot below shows the output voltage as compared to the control PWM cycle.

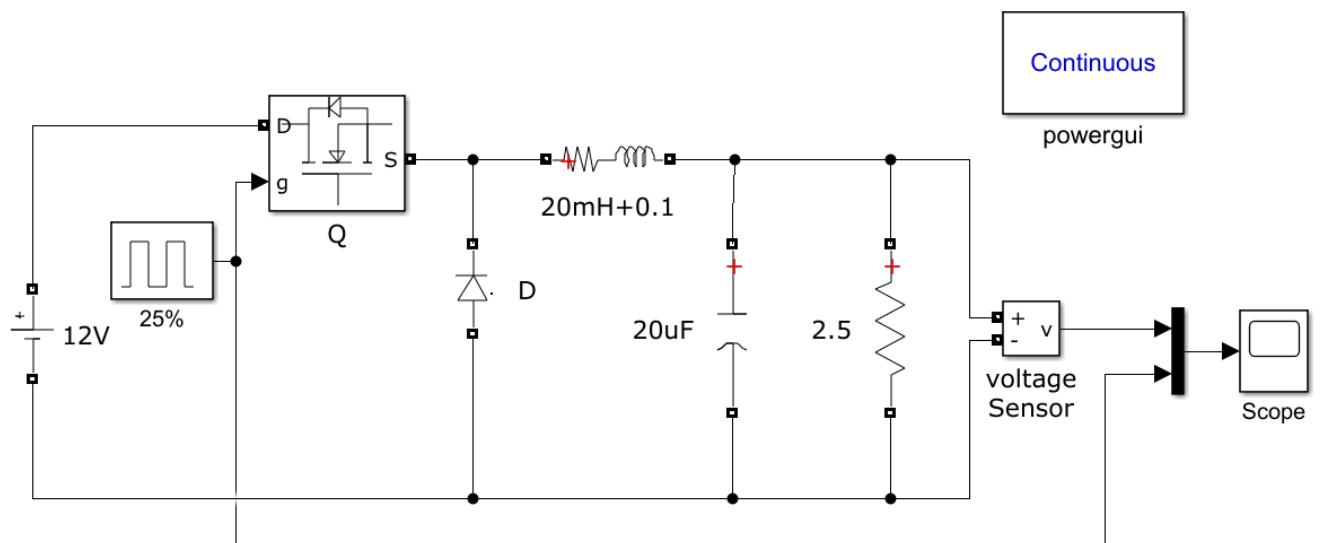
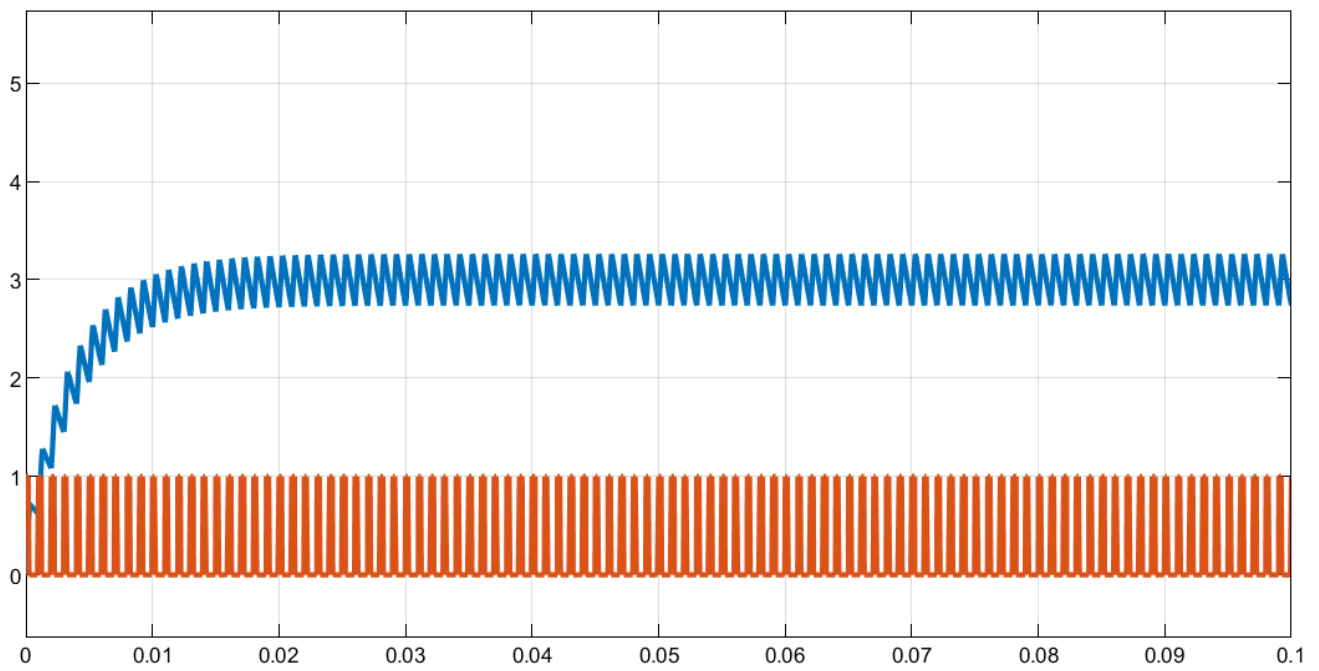


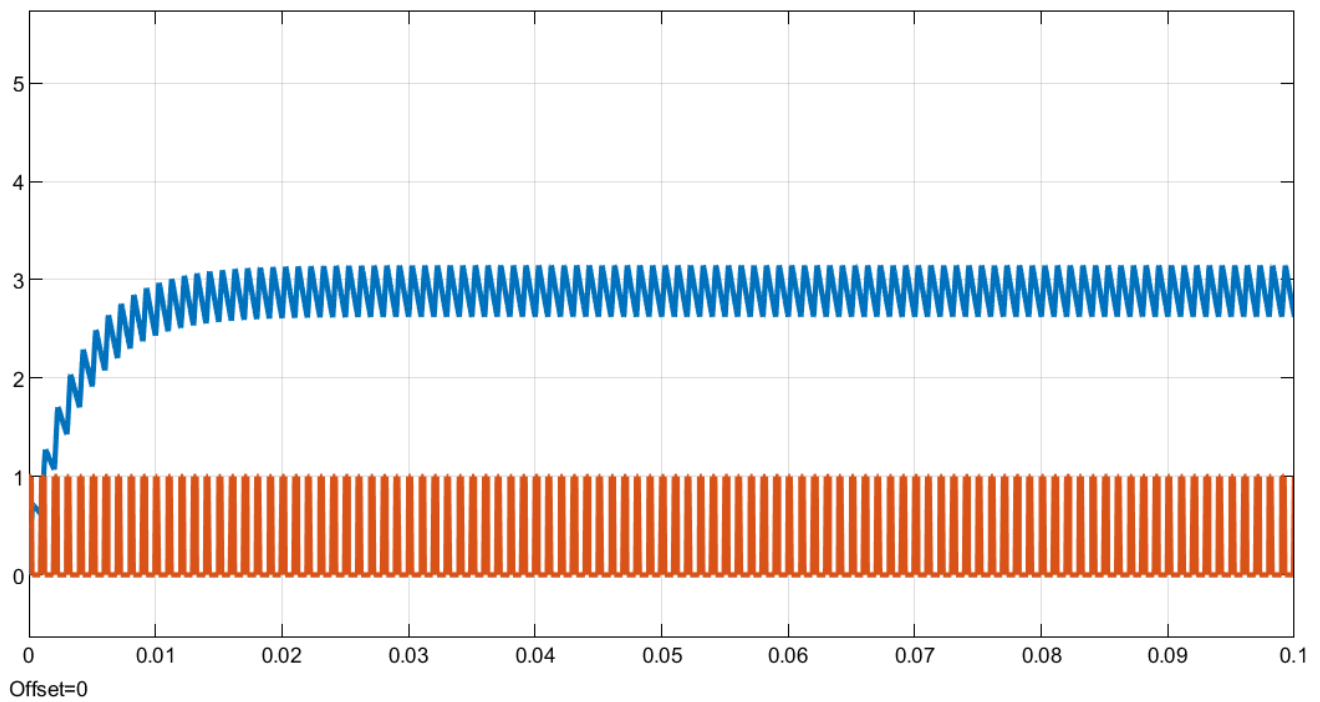
12) There are copper loss in inductor.



13) Change the duty cycle of PWM to 25%.







Marking scheme

Each lab session will be assessed and will count towards the overall mark for the practical component of this course. The lab demonstrator will check your measurements and the quality of presentation of data in the lab book and plots.

Laboratory report (100/100 marks)

- Objective & background (total mark - 30/100)
- Calculations, graphs, and results (total mark - 40/100)
- Discussion of results and conclusions (total mark - 30/100)